ModelIt!

Student Activity Pack

Can a Robot Swim Like a Beaver?

How Nature Inspires Engineering Design

Grade NE | NGSS 4-LS1-1, 3-5-ETS1-1, 3-5-ETS1-3 | Structure and Function

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What animals do you think are the best swimmers? Why?**

2. **If you were designing a robot that needed to swim, what animal would you copy?**

3. **Draw what you think the perfect swimming tail looks like.**

4. **What does the word 'tradeoff' mean to you?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Tail Shape | Body Size | Swimming Speed | Energy Used

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Beaver Tail

Flat tail, medium body

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Round Tail

Round tail, medium body

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Big vs. Small

Flat tail, changing body size

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Tail Flexibility — How stiff or bendy the tail is — flexibility affects how water flows around it

Water Temperature — Cold water is thicker than warm water, which changes how easy it is to swim

Surface Texture — Whether the tail surface is smooth, rough, or has scales

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do engineers study animals when they want to build better machines? Use evidence from your simulation to support your answers.

1. **Why is a flat tail better for swimming than a round one?**

2. **What is the tradeoff between body size and energy use?**

3. **Name another animal with a body part that engineers might want to copy. Why?**

4. **Why do engineers study nature instead of just inventing everything from scratch?**

5. **How did the model help you make better design choices for your paddle?**

8 STEM Challenge: Paddle Design Challenge

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YOUR ENGINEERING MISSION

Design a paddle inspired by a beaver tail that moves a toy boat the farthest distance across a water bin in one push.

THE SCENARIO

A toy company wants to make the best paddle-powered boat. Your engineering team has been asked to test different paddle shapes inspired by animal tails. Use your model data to make your design choices!

Engineering Design Thinking

1. WHAT SHAPE WILL YOU MAKE YOUR PADDLE — FLAT, CURVED, CUPPED, OR SOMETHING ELSE?

2. WHAT SIZE SHOULD YOUR PADDLE BE?

3. WHAT MATERIAL WILL YOU USE — WILL IT BE STIFF OR FLEXIBLE?

4. HOW WILL YOU MEASURE WHICH PADDLE WORKS BEST?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Biomechanical Engineers study how animals move and use those ideas to design better robots, prosthetics, and vehicles. They earn $70,000–$120,000/year.